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A dynamic size-structured population model: does disturbance control size structure of a population of the massive coral *Gardineroseris planulata* in the Eastern Pacific?

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Abstract Using a long-term data set of *Gardineroseris planulata* (Dana) on Uva Island reef, Panamá, we developed a simulation model that relates size-specific schedules of growth and partial mortality to predation by *Acanthaster planci* (Linnaeus) and El Niño–Southern Oscillation (ENSO)-related elevation of water temperature. We compared the model predictions to field observations for both the subpopulation of colonies that was used for model development and for the entire population. No statistically significant differences in the size-frequency distributions of the real and modeled coral populations were found for the subpopulation during any of 9 yr or for the entire population during 4 yr. These results suggested that the model relationships were reflecting field conditions. Longer-term (100 yr) simulations were conducted to assess the relative importance of predation and ENSO-related colony losses in determining the size structure of the coral population. Predation by *A. planci* was of overwhelming importance due to both stronger effects of predation (larger transitions) and the frequency of predation (yearly) compared to ENSO (episodically). Even the least-frequent predation scenario skewed the distribution toward smaller colonies, while simulations where populations were subjected to frequent ENSO events (≥ 3 yr) still maintained colonies in the largest size class. The model results suggested that sea star predators may not have been in the present abundance on this reef prior to the last 30 yr; with predators present, the model predicts that the distribution would be skewed toward smaller colonies.

Introduction

Recent evidence has demonstrated that natural disturbance is an important factor affecting coral populations, community dynamics, and ecosystem structure and function (e.g. Connell 1978; Connell and Keough 1985; Hughes and Jackson 1985; Huston 1985; Glynn and Colgan 1989; Knowlton et al. 1990; Done 1992; Porter and Meier 1992; Witman 1992). Several kinds of biotic and physical disturbances in tropical, near-shore, marine environments (e.g. sea-level changes, storms, coral bleaching events, predator outbreaks), of varying intensity and time scales, have been well documented (e.g. Johannes 1975; Endean 1976; Knowlton et al. 1981; Pearson 1981; Potts 1983; Hughes and Jackson 1985; Stoddart 1985; Williams and Bunkley-Williams 1990; Glynn 1993). However, because the time scales of both disturbance and recovery processes are long (Colgan 1981, 1987; Glynn 1990), few studies have data from pre-disturbance through recovery to the original size structure of coral populations. Thus, our knowledge of the effects of disturbances is largely fragmentary (Done 1992; Porter and Meier 1992).

One approach to understanding coral population-dynamics uses demographic models to relate short-term population processes to longer-term changes in the size structure of the population; these models are based on work with terrestrial plants (Usher 1966; Hartshorn 1975; Werner and Caswell 1977; Enright and Ogden 1979). Hughes (1984) was the first to develop a demographic model of coral populations by modifying a traditional size or stage-structured matrix model (Usher 1972; Werner and Caswell 1977) to include transitions unique to clonal organisms such as colony fusion, fragmentation, and shrinkage (Caswell 1985: review; Hughes and Jackson 1985). Other factors included in Hughes' (1984) coral model were indeterminate growth and density-independent recruitment rates for an "open" population with planktonic larvae. Acknowledging the importance of disturbance to coral popula-

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tions, Hughes developed two matrices, one for use in calm years and the other in stormy years. Similarly, Done (1987) substituted a "damage" matrix to simulate partial and complete mortality of coral colonies ascribed to the predatory sea star *Acanthaster planci* (Linnaeus) in years when these predators were in outbreak abundances. In the last 12 yr, these modeling techniques have been used successfully with several clonal organisms, including corals (Hughes and Jackson 1985) and bryozoans (Hughes 1990).

Transition matrices used in population models are composed of probabilities that a coral colony will move from one size class to another based on field data of size-dependent growth and shrinkage of colonies. One of the underlying assumptions of previous population models for corals was that within calm years or years with low abundances of predators, transition rates between size classes due to growth and mortality were relatively constant. This assumption was justifiable in areas subjected to outbreaks of *Acanthaster planci* that devastated whole populations (Done 1987), or where calm years were interrupted by major hurricanes with similarly severe effects on coral populations (Hughes 1984). These models used the mean transition rate for growth and shrinkage calculated for all the years of calm and low predator abundance for "normal" model years without disturbance. In the eastern Pacific, however, although *A. planci* is an important cause of coral mortality (Glynn 1974), it has not been reported in outbreak abundances and its effect on corals is less devastating than in the western Pacific (Glynn 1974). In addition, some areas in the eastern Pacific are subjected to sea-surface warming during El Niño-Southern Oscillation (ENSO) and cooling due to thermocline shoaling or upwelling (Glynn 1990). Such thermal variability affects coral growth and mortality, independent of the number of predators. In our model, we incorporated these two factors by varying the transition matrix yearly as a function of the abundance of *A. planci* and the occurrence of ENSO. These factors have been demonstrated to be of great importance to coral populations in the Eastern Pacific (Glynn and Stewart 1973; Glynn 1974, 1985, 1988, 1990, 1993; Glynn et al. 1989).

This paper is composed of three parts. First, we developed a size-structured matrix model for coral whereby the transition rates representing growth and partial mortality were temporally variable. Next, the model was tested by comparing model predictions with independently obtained field data of actual population size-structure. Finally, several scenarios were used to explore the long-term effects of both intermittent and ongoing disturbances in controlling the size structure of the population. These long-term scenarios lasted 100 yr, as this was sufficient time for trends in population size-structure to appear under the modeled conditions. We chose to model *Gardineroseris planulata* (Dana), an uncommon yet widespread species of massive coral (Veron and Marsh 1988; Veron 1992) found in the eastern Pacific. We were interested in comparing the effects of two

different disturbances, predation by *Acanthaster planci* and ENSO-related mortality. Glynn (unpublished data) had detailed records on the size structure of one subpopulation of this species for 11 yr covering a severe ENSO event and predation by the sea star *A. planci*.

Materials and methods

Field populations

Colonies of *Gardineroseris planulata* (Dana) on Uva Island reef, Panamá were counted and measured periodically from 1970 to 1994 (present Fig. 1; Glynn 1985, 1990, and unpublished data). We measured colony area by laying a flexible screen ($\sim 1 \text{ m}^2$ with mesh size of 4 cm^2) over each colony and counting the number of grids that contained coral tissue. Colonies used to develop parameters for the model were a subpopulation ($n = 23$) in one area of the reef called *Gardineroseris* City that occupied $\sim 32 \text{ m}^2$ of reef bottom. The size of each colony in the subpopulation was measured in 1970, 1980, and yearly from 1984 to 1994 (Fig. 1). For modeling purposes, colony area was transformed to colony diameter, assuming that each colony approximated a hemisphere. This was a reasonable assumption as a regression of diameter and height (= radius) of colonies during the 24 yr they were measured resulted in a slope of 1.9 ($r^2 = 0.78$); a slope of 2.0 would represent a perfect hemisphere (Glynn unpublished data).

Gardineroseris planulata growth over the 24 yr period was calculated from field data, assuming growth to be constant over the 10 yr period from 1970 to 1980, when no major ENSO events occurred. To complement the field data, an additional estimate of growth was obtained from the largest colony in *Gardineroseris* City. The colony was cored vertically for 254 cm, exposing a continuous coral skeletal record of 192 yr. Distances between annual skeletal bands were measured along three transects in the top 50 cm of the core and used to calculate the mean and range of year-to-year change in colony size as described by Glynn (1985).

Acanthaster planci (Linnaeus) occupying the reef were counted by daytime surveys of the entire reef yearly from 1975 to 1994 (Fig. 1) using SCUBA; these sea stars are usually exposed, so no effort was made to check under loose rubble. From 1970 to 1983, the coral colonies within *Gardineroseris* City that were used for model development were protected from sea star predation by a natural biotic barrier formed by *Pocillopora* spp. *A. planci* does not cross continuous stands of *Pocillopora* spp. because of the coral's nematocysts and symbiotic crustacean guards (Glynn 1985). The *Pocillopora* spp. barrier was eliminated in 1983 when prolonged sea-warming killed the protective corals, and thus counts of

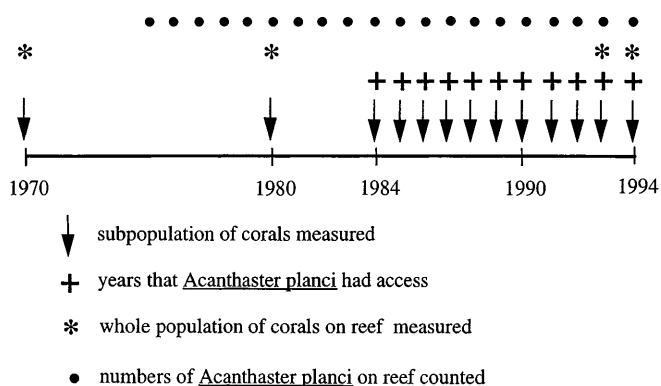


Fig. 1 Time-line of events and sampling of *Gardineroseris planulata* at *Gardineroseris* City, Uva Island study reef, Panamá.

A. planci were related to shrinkage of colonies in *Gardineroseris* City only when *A. planci* had access to the colonies (1984 to 1994).

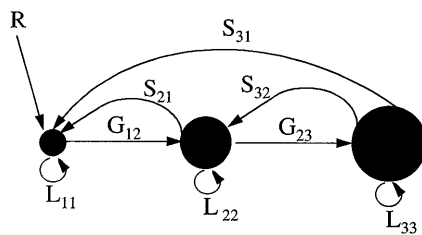
No sexual recruitment from planktonic larvae to the *Gardineroseris* City subpopulation was observed during the 24 yr of the study; however, sexual recruitment to the rest of the population was estimated in March 1994 (Glynn unpublished data) from colonies in the smallest size class (0 to 20 cm). Small colonies with a hemispherical, undamaged appearance were divided into “yearly” classes based on an average growth in diameter of 2.6 cm yr⁻¹, estimated from the skeletal record of one large colony. We reasoned that these small colonies might have a tight relationship between size and age and therefore could be used to hindcast recruitment.

General model

A size-class matrix-projection model similar to those developed by Hughes (1984) and Done (1987) predicts the size structure of a population at time $t + 1$ from the size structure at time t and a transition matrix, $\bar{A}$. The form of the model is:

$$X_{(t+1)} = \bar{A}X_{(t)} + \begin{bmatrix} R \\ 0 \\ 0 \end{bmatrix}, \quad (1)$$

where $x_{(t)}$ is a column vector describing the population size-structure at time t , and $\bar{A}$ is a transition matrix that determines the dynamics of the population (Fig. 2 is a graphical representation of a general matrix model for a population with three distinct size classes). Changes in population size structure through time are simulated by iteration. Each year, a colony can either stay in the same size class (L), grow one size class (G), or shrink to any of the smaller size classes (S). In addition, “new” colonies can recruit to the smallest size class from planktonic larvae (R). In the 24 yr of field data, complete mortality of a colony was never observed, so we did not include it in the model.



$$X_{(t+1)} = \bar{A} X_{(t)} + \begin{bmatrix} R \\ 0 \\ 0 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} L_{11} & S_{21} & S_{31} \\ G_{12} & L_{22} & S_{32} \\ 0 & G_{23} & L_{33} \end{bmatrix}$$

Fig. 2 General matrix model for three size-class model; our model has 11 size classes (● coral colonies within different size classes; L loops, representing proportion of colonies that remain within a size class; G growth, proportion of colonies that grow into next largest size class; S shrinking, proportion of colonies that suffer partial mortality and shrink into any of smaller size classes; R recruitment, number of colonies that recruit from planktonic larvae [Subscripts = i (rows) and j (columns) designations for matrix model; for further explanation of model terms see “Materials and methods” – General model]

In the present model, the population of *Gardineroseris planulata* was divided into 11 size classes based on colony diameter. We used diameter rather than area for this coral model to make our results comparable to an earlier model assessing the impact of predation by *Acanthaster planci* on massive corals (Done 1987, 1988). The first ten size classes represented increments of 20 cm, while the largest size class included all colonies >200 cm in diameter. The matrix was composed of the probabilities of each transition (L_{ij} , G_{ij} , S_{ij}) for colonies within each size class (Fig. 2). Probabilities of transitions varied from year to year, depending on the field population’s response to ENSO and predation by sea stars.

Model development

Transition rates due to growth (G)

We calculated G from the transitions of colonies in the *Gardineroseris* City population from one size class to the next larger size class. Smaller corals grew more rapidly than large corals (Fig. 3). According to these estimates of growth, >25% of the corals in the two smallest size classes (≤ 40 cm) moved into the next largest size class every year. The next two size classes (41 to 80 cm diam) had lower transition rates than the smallest colonies. Average G based on field data was 11% in size classes >80 cm.

Mean linear growth from the skeletal record was 1.32 cm yr⁻¹, with a range of 1.2 to 1.4 cm yr⁻¹. Skeletal cores give radial growth, so the range of yearly change in diameter was 2.4 to 2.8 cm yr⁻¹. Measuring growth in the top 50 cm of this colony means that the colony was 150 cm in diameter at the innermost (youngest) measurement. Thus, we compared the skeletal data to the estimate derived for growth of corals in the larger size classes (>80 cm diam) from the field data. Assuming that corals were evenly distributed within a size class, the fraction of corals that “graduate” into a higher size class each year was calculated as growth divided by the width of the size class. Based on the skeletal data, G for colonies >80 cm in diameter ranged from 0.12 to 0.14 (or 12 to 14%). Thus, the skeletal record approximated our estimate of growth for larger colonies derived from field data (11%).

For use in the model, we divided the log curve of growth (Fig. 3) into three sections and averaged the transitions within these sections. Thus, our baseline values for G for colonies with diameters of 0 to 40 cm = 0.24, 41 to 80 cm = 0.14, and >80 cm = 0.11.

Transition rates due to partial mortality (S)

Transitions for partial mortality (S) were calculated from the field population and are assumed to be the additive effect of predation by *Acanthaster planci* and ENSO-associated mortality. We used the additive relationship between these two sources of mortality

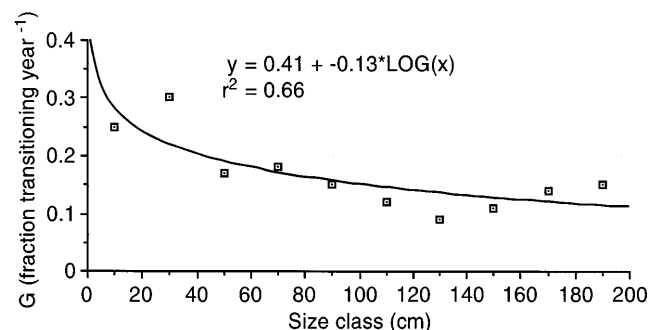


Fig. 3 *Gardineroseris planulata*. Size-dependent transition rates due to growth (G) of colonies at *Gardineroseris* City: a summary of transitions from smaller to larger size classes for subpopulation from 1983 to 1993

because it was the simplest to construct and we reasoned that the proportion of a colony that died due to ENSO was independent of the proportion removed by the predators. For the *Gardineroseris* City subpopulation, predation by *Acanthaster planci* and ENSO mortality were temporally distinct events.

Mortality due to predation by *Acanthaster planci*. To quantify transition rates due to the predator, we first quantified the abundance of *A. planci* and mortality rates of *Gardineroseris planulata* in the *Gardineroseris* City subpopulation on Uva Island. From 1984 to 1994, the number of *A. planci* ranged from 0 to 12 sea stars ha^{-1} , with a mean of 4.3 sea stars ha^{-1} (Fig. 4a; and Glynn 1990). By 1987, after 4 yr of exposure to predation, >45% of the surface area covered by colonies of *G. planulata* in *Gardineroseris* City had been removed by sea stars (Fig. 4a).

Mortality rate was then related to the number of *Acanthaster planci* present on the reef during non-ENSO years, after the protective biotic barrier of *Pocillopora* spp. had been removed. There was a significant linear relationship between the number of sea stars ha^{-1} and the yearly change in area covered by *Gardineroseris* City colonies from 1984 to 1994 (Fig. 4b). Although predators consumed, *G. planulata* whenever they were present on the reef (Glynn personal observations) when there were <3 predators ha^{-1} growth of coral balanced or surpassed removal of tissue by predators and there was a net increase in the area covered by this coral (a decrease in percent dead in Fig. 4a). Thus, at very low abundances, mortality did not exceed the cumulative growth rate of all the colonies (Fig. 4b). In contrast, when predator abundances exceeded 3 sea stars ha^{-1} , there was usually a net loss of area covered by *G. planulata*.

We investigated the predators' size preference of prey by quantifying the sequence of attacks by sea stars, the number of

attacks within each size class, and the amount of tissue removed from each colony each year. In all 8 years that *Acanthaster planci* attacked colonies in *Gardineroseris* City, there were significant differences between the size-frequency distribution of colonies within the subpopulation and the size-frequency distribution of colonies that were attacked (Komogorov-Smirnov test for goodness of fit; $p < 0.05$ for all eight tests). In 1984, three of four colonies attacked had diameters that ranged from 40 to 80 cm. In each case, enough tissue was removed from the colony to result in a transition into the next smaller size class. In 1985, the same colonies were attacked again, further reducing their size class. Five of the six additional colonies attacked in 1985 (83%) were within the 40 to 80 cm size range. By 1987, all the colonies within the 40 to 80 cm diam size-range had been attacked and reduced in size. Although >95% of the coral tissue was removed from some colonies, there were no cases where there was complete mortality. During this same year, *A. planci* began to attack larger colonies outside of the "preferred" size range. Colonies in *Gardineroseris* City < 40 cm in diameter were never attacked.

To parameterize the model, we estimated transition rates due to predation from field data of the size of *Gardineroseris planulata* colonies in *Gardineroseris* City from 1984 to 1994. In this time interval all mortality was assumed to be due to predation. Each year, we counted number of attacks and measured the area of tissue removed from individual coral colonies. Changes in area were transformed to changes in diameter using the assumptions of Done (1987, 1988), who also modeled predation rates of *Acanthaster planci*. All occasions when an individual shrank one or more size class were tallied and transition rates were calculated.

A rectangular hyperbola in the form:

$$S = \frac{S_{\max} \cdot \text{sea stars}}{\text{sea stars} + K_s} \quad (2)$$

was fitted to the non-linear relationship between abundance of predators and the transition rates for each size class (Fig. 5). In this equation, S = the transition rate used in the matrix model for each yearly iteration, sea stars = the average abundance of sea stars on the reef each year, $S_{\max}$ = maximum transition rate found in the field, and K_s = half saturation constant, which determines the shape of the curve and represents the number of sea stars that will produce half the maximum shrinking rate. Values for each parameter for each transition rate curve are given in Table 1.

For each model run, the number of predators for each year of simulation was input, and transitions due to mortality were calculated with this equation.

To obtain an estimate of the fit between the field data and the equation used in the model, we solved the equation for K_s :

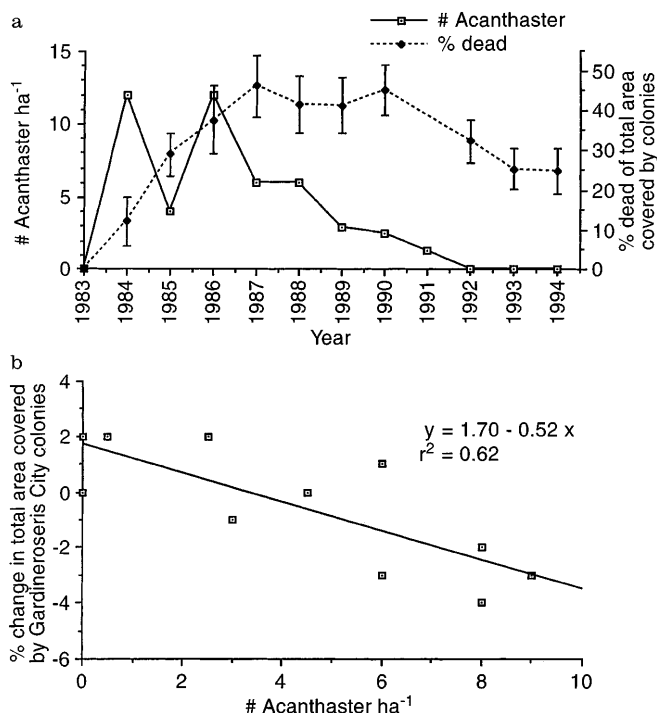


Fig. 4 a *Acanthaster planci* abundance and *Gardineroseris planulata* mortality. Density of predator *A. planci* on Uva Island reef from 1983 to 1994; during this period, coral colonies at *Gardineroseris* City were exposed to predation following death of *Pocillopora* spp. barrier; also shown is mean (bars are $\pm$ SE) percentage of each colony that was dead. b Linear regression between number of *A. planci* and partial mortality rate of *G. planulata* at Uva Island reef

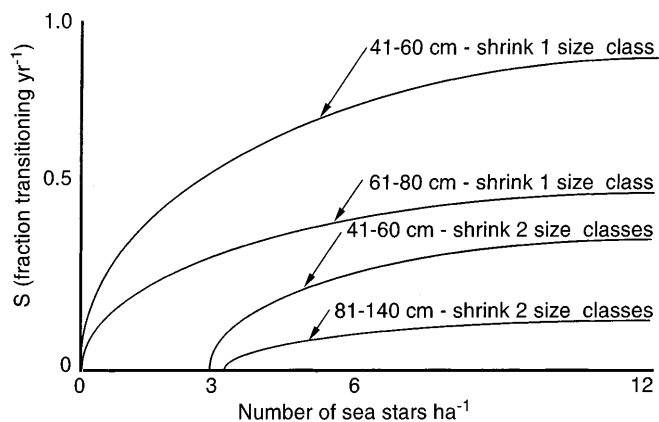


Fig. 5 *Acanthaster planci* and *Gardineroseris planulata*. Relationships between density of predator and fraction of coral colonies that shrink (S) to smaller size classes. Parameter values for all relationships are listed in Table 1

Table 1 *Acanthaster planci* and *Gardineroseris planulata*. Parameter values for model equations (Eqs. 2 and 3) relating density of predator *A. planci* to partial mortality rates of each size class of *G. planulata* colonies. Mortality due to *A. planci* is modeled as a rectangular hyperbola for each size class (*x-intercept* density of seastars at which colonies in a given size range are attacked and

tissue area is reduced by a given number of size classes; $S_{\max}$ mortality rate (%) above which seastar predation on a size class is “saturated” and predators select colonies in a less preferred size class; K_s half-saturation constant – number of *A. planci* that produces half maximum shrinking rate)

Transition rate (from → to)	Parameters for non-linear equation relating mortality to <i>A. planci</i> density			
	<i>x-intercept</i>	$S_{\max}$	K_s	r^2
2→1	1	38	1.5	0.65
3→2	0	80	1.0	0.88
3→1	3	30	3.5	0.62
4→3	0	45	1.0	0.79
4→2	3	20	3.1	0.56
4→1	6	15	6.2	0.43
5,6,7→4,5,6 (shrink one size class)	3	15	3.2	0.87
5,6,7→3,4,5 (shrink two size classes)	4	7	4.1	0.75
8,9,10,11→7,8,9,10	3	10	3.1	0.33

$$K_s = \frac{\text{sea stars } (S_{\max} - S)}{S} \quad (3)$$

This resulted in a linear equation and we used a least-squares approach to estimate how well the equation used in the model fit the estimate of K_s calculated from the field data. Values for r^2 ranged from 0.33 to 0.88, with the higher values for the most frequent transitions (Table 1).

Mortality due to ENSO. Partial mortalities of colonies in *Gardineroseris* City in 1982–1983 were attributed to high and prolonged sea-surface temperatures (SST) that accompanied ENSO. *G. planulata* colonies at Uva Island were relatively undisturbed by the severe ENSO; only 13% of the *Gardineroseris* City colonies suffered any degree of partial mortality (Fig. 6). Field data on loss of coral tissue for each colony during the ENSO was grouped into size classes, and transition rates from one size class to another tallied. Thus, we used empirical, system-specific transitions for ENSO mortality rates measured at Uva Island. In each run of the model, for every year that ENSO occurred, transition rates due to ENSO were added to transition rates due to predation to form values for S in the transition matrix.

Model formulations describing recruitment rates (R)

Number of recruits to the Uva Island reef population ranged from 1 to 11 yr⁻¹ (Fig. 7). We modeled sexual recruitment from planktonic larvae as a stochastic function by drawing the number of recruits yr⁻¹ randomly from a uniform distribution ranging from 1 to 11. When drawing numbers from a uniform distribution, there is

equal probability that each number will be chosen. Because the field data did not conform to a normal distribution and showed no other apparent pattern, we felt that a uniform distribution was an adequate choice. In the model, recruits were added to the smallest size class for each yearly iteration.

Forcing function formulations: *Acanthaster planci* and ENSO

The model had two forcing functions: (1) density of *Acanthaster planci* caused partial mortality, and (2) frequency and intensity of ENSO caused cessation of growth as well as partial mortality. Values for these functions were input for each year of simulation. The density of *A. planci* was taken from empirical data quantifying the yearly average number of sea stars ha⁻¹ over the period to be simulated (data taken from Fig. 4a).

The model formulation for ENSO was more complex. The frequency and intensity of ENSO events were derived from historical data (Glynn 1988). In the model, we developed an “intensity” scalar for ENSO that was based on the number of days each year that the temperature exceeded 30 and 32 °C. Glynn and D’Croz (1990) found that *Pocillopora damicornis*, a major reef-building coral in the tropical eastern Pacific, suffered significant bleaching and subsequent partial mortality after being placed in experimental units subjected to 32 °C for 2 wk. The same level of mortality occurred after 8 wk at 30 °C. Thus, a warming event with

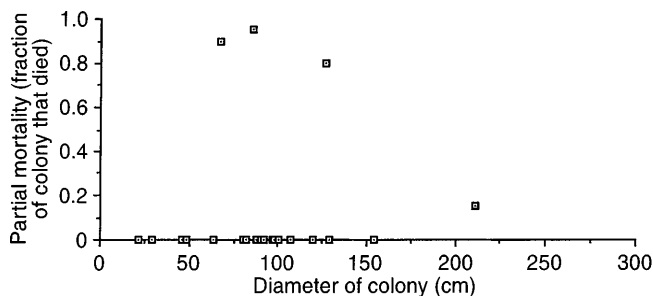


Fig. 6 *Gardineroseris planulata*. Size-dependent partial mortality of colonies in *Gardineroseris* City during 1982–1983 ENSO event

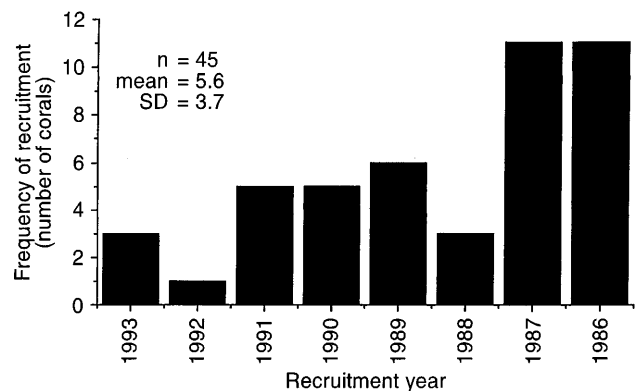


Fig. 7 *Gardineroseris planulata*. Estimated number of colonies that recruited to Uva Island reef from 1986 to 1993. Recruitment per year was calculated by assuming constant yearly growth rate and no partial mortality

a maximum temperature of 32 °C was four times as “intense” as one with a maximum temperature of 30 °C. The following quantity incorporates both duration and magnitude of warming events:

$$\text{ENSO intensity} = (\text{number of days} \geq 32^\circ\text{C}) + \left(\frac{\text{number of days} \geq 30^\circ\text{C and} < 32^\circ\text{C}}{4} \right) \cdot (4) \quad (4)$$

Glynn and D'Croz (1990) found that *Pocillopora damicornis* bleached and suffered mortality under conditions when the ENSO intensity scalar was ≥ 14 . Observations at Uva Island suggested that *Gardineroseris planulata* was less sensitive to bleaching and mortality due to extreme high temperatures than was *P. damicornis*. Thus, in our model, *G. planulata* was resistant to extreme high temperatures until the scalar exceeded 21; clearly, this choice of scalar needs to be tested experimentally. In the real world, this assumption translated into mortality for *G. planulata* subjected to 21 d above 32 °C, 84 d above 30 °C, or any linear combination between.

Testing the model: conditions of simulations

Does model output match Gardineroseris planulata population dynamics?

We quantified how well the model matched change in the size structure of the *Gardineroseris* City subpopulation on which it was based. The model was run under the conditions of *Acanthaster planci* density and ENSO intensity found for the subpopulation for each year from 1980 to 1994. We set the initial size structure equal to the size structure of the field population in 1980, then ran the model for 14 yr. There were 9 yr of field data for the *Gardineroseris* City subpopulation during this time. We compared size-frequency distributions predicted by the model to those measured in the field using a Komogorov–Smirnov test for goodness of fit.

Next, size data for the entire population of *Gardineroseris planulata* on Uva Island were compared to output of the simulation model. The simulation was begun with the initial size structure of the coral population in 1970, and run for 24 yr under conditions of predation and ENSO measured in the field. Thus, each year we input the average yearly density of *Acanthaster planci* as reported by Glynn (1990). An ENSO was simulated in Year 13 of the model run, to correspond to the 1982–1983 ENSO event that occurred in the eastern Pacific. The objective of this simulation was to isolate the effects of predation and ENSO mortality without the confounding effects of recruitment. Thus, for this experiment, we ran the model with no recruitment, to track what occurred to the existing population on the Uva Island reef.

Testing recruitment portion of the model

To test the recruitment part of the model, the model was run from 1970 to 1994 exactly as described earlier, but stochastic recruitment was added. Because the stochastic function added variability to the output, the simulation was repeated 25 times, and means were calculated for predictions of number of colonies in each size class. The data that we tested this simulation against included all colonies ≤ 20 cm diam.

Comparing different disturbances

To predict the longer-term effect of each disturbance (i.e. predation by *Acanthaster planci* and ENSO-associated mortality) on the size structure of a population of *Gardineroseris planulata*, we ran the model several times under the initial conditions (1970) found at the Uva Island reef, but with differing disturbance regimes. The duration of each simulation was 100 yr. First, we varied predator abundance and ENSO frequency independently, and then varied the two disturbances together. Last, we estimated time to recovery, defined as a return to the original size structure, for the disturbed Uva Island reef population if disturbances were to cease completely in 1994.

Predation scenarios. The predation scenarios were based on the patterns of abundance of *Acanthaster planci* found at the Uva Island reef from 1970 to 1982 (Glynn 1990). However, along with using the actual field values, we ran some alternative scenarios to determine the effect of differing duration of predator abundance. In the field, predator abundances varied from 0 to 12 sea stars ha^{-1} ; the pattern of abundance was somewhat cyclical, with 10 to 12 yr between minimum abundances. For this series of simulations, *A. planci* abundance was modeled as a sinusoidal curve, with a range of 0 to 12; periods were 8, 12, and 20 yr. These periods were chosen to model the field data (12 yr) and then to investigate the effects of shorter and longer periods.

ENSO scenarios. The object in varying the frequency of strong to very strong ENSO events was to determine the long-term effect of repeated warming events on the size structure of a *Gardineroseris planulata* population. In the first simulation, historical data of the frequency of ENSO events from 1883 to 1983 were used (Glynn 1988). In this model run, strong to very strong ENSO events were simulated in Years 4, 7, 29, 34, 42, 47, 57, 72, 79, 91, and 98 of the simulation. Two additional ENSO scenarios were developed from the cumulative probability distributions that indicated the likelihood of ENSO occurrences (Glynn 1988). In one of the two alternative scenarios, intervals between ENSOs were selected from a uniform distribution ranging from 0 to 24; this distribution covered the range of frequencies estimated by Glynn. In the other, intervals were selected from a normal distribution, with the mean interval between ENSO events = 11.95 (S.D. = 6.60); this was the mean recurrence time for strong events calculated by Glynn. More scenarios were run to determine the ENSO frequency necessary to cause a long-term effect on the population size-structure.

Combining ENSO and predation scenarios. Simulations combining the historical frequency of ENSO and all three predation scenarios were run.

Investigating population recovery

The object of this model scenario was to estimate the post-disturbance resilience of the *Gardineroseris planulata* population at Uva Island reef. As in Done (1987), resilience was defined as the time needed for the coral population to regain the pre-disturbance size structure in the absence of any subsequent disturbance. To accomplish this objective, the model was run under the Uva Island conditions from 1970 to 1994, and then run for the next 76 yr with both predation and ENSO set to zero.

Results

Does model output match Gardineroseris planulata population dynamics?

We accepted the null hypothesis that there were no statistically significant differences between the size-frequency distributions predicted by the model for the *Gardineroseris planulata* population in *Gardineroseris* City and the distributions measured in the field for all 9 yr compared ($p > 0.05$).

During the 24 yr period for which we compared model predictions to field measurements of the size structure of the whole population, the size structure of the coral population varied greatly (Fig. 8). In 1970, the field population was nearly normally distributed, with

the greatest number of corals falling within the 80 to 100 cm size class (Fig. 8a). From 1970 to 1980, the reef on Uva Island was subjected to predation from *Acanthaster planci*, but to no ENSO events. By 1980, after 10 yr of predation, the structure of the *Gardineroseris planulata* population had changed dramatically; most of the colonies were <40 cm in diameter (Fig. 8b). There were no significant differences between the size-frequency distributions predicted by the model and the distributions measured in the field (Komogorov–Smirnov test for goodness of fit, $p > 0.05$). There was a slight, but not significant, underestimation of the amount of damage to coral colonies in the 41 to 60 cm size class. This discrepancy resulted in the model predicting more corals in this size class, and less in the 21 to 40 cm size class, than found in the field. However, the model accurately predicted the elimination of colonies in the larger size classes, and the accumulation of remnant colonies in the smallest size class that occurred in the field.

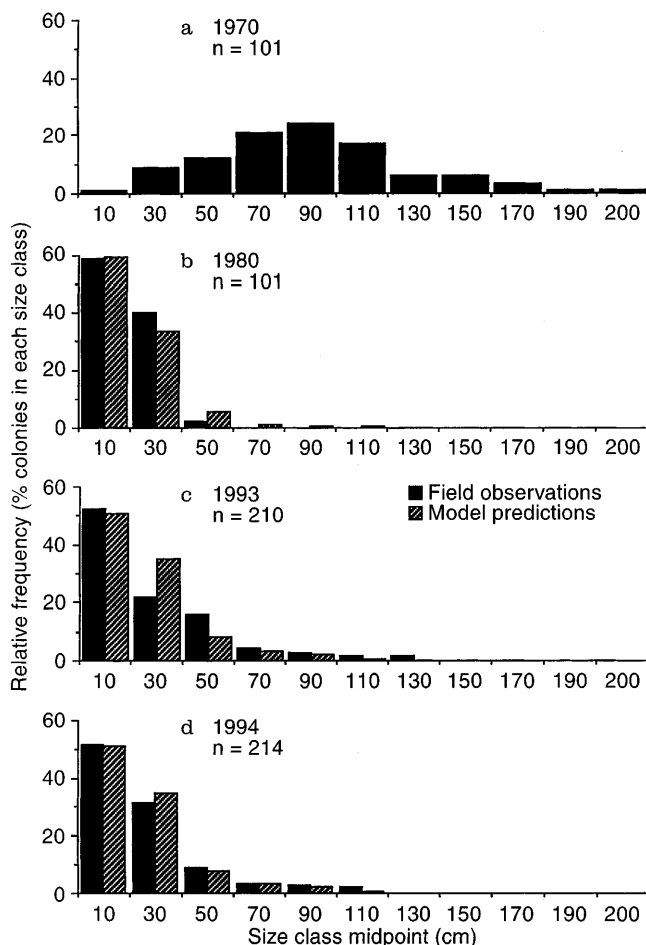


Fig. 8 *Gardineroseris planulata*. **a** Size structure of entire population on Uva Island reef in 1970, used as initial condition in this simulation; **b–d** field observations and model predictions of relative frequency of coral colonies in each size class for 24 yr model-run simulating total population under conditions at Uva Island reef

From 1980 to 1993, both of the processes driving partial mortality of the coral colonies occurred. Predation was higher in the early years of the simulation, declined through time, and was absent from 1992 to 1994 (Fig. 4a). In 1982/1983 there was a very strong ENSO event. By 1993, the corals began to recover from this series of disturbances as some colonies entered the larger size classes (Fig. 8c). Model predictions reflected this initial recovery. The model predictions matched the relative abundance of colonies observed to fall within the smallest size class. From 1980 to 1993, the relative abundance of colonies in the 0 to 20 cm size class declined from ~60% to just over 50% in both the field and model populations. However, the model underestimated the growth rate from the 21 to 40 cm into the 41 to 60 cm size class; this error resulted in a prediction that overestimated the number of corals in the 21 to 40 cm size class and underestimated the number of corals in the 41 to 60 cm size class. In contrast, the model accurately predicted the increase in relative abundance in the next three larger size classes.

From 1993 to 1994, the relative abundance of colonies in the smallest size class remained about the same. There was an increase in the number of colonies in the 21 to 40 cm size class and a decrease in the number of colonies in the 41 to 60 cm size class observed in the field population (Fig. 8c, d), suggesting that there was some mortality in the 41 to 60 cm size class. Although this mortality was unaccounted for in the simulation model, the slowing of recovery in the field population resulted in a better fit between the model predictions and the field observations in 1994. Again, there were no significant differences detected between the size-frequency distributions predicted from the model and measured in the field (Komogorov–Smirnov test for goodness of fit, $p > 0.05$).

Testing recruitment

We compared the number of colonies in the smallest size class (0 to 20 cm) observed on the reef to the number predicted by the model to test the recruitment part of the model (Fig. 9). For all 4 yr quantified, there were no statistically significant differences in the number of colonies predicted by the model and those observed in the field (chi-square test, $p > 0.05$). Variances were small; SE of five repeated simulations ranged from 3 or 4% of the mean values. When simulations were replicated 25 times, variances did not decrease.

Comparing different disturbances

Predation scenarios

In all the model scenarios varying predation (Fig. 10), the model predicted that most of the coral colonies would be small. When predation was modeled on an 8 yr

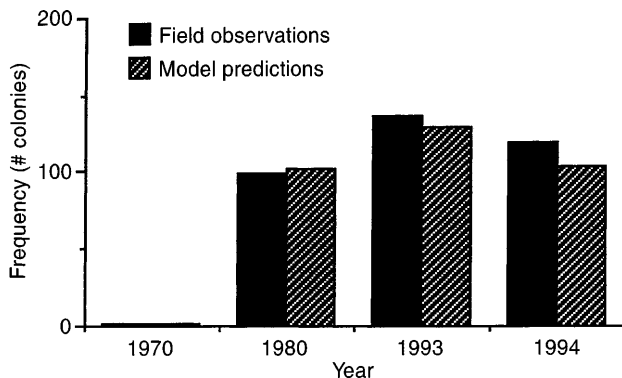


Fig. 9 *Gardineroseris planulata*. Field observations and model predictions of number of colonies ≤ 20 cm in diameter when stochastic recruitment was added to model. Standard errors are too small to include in graph (= 3.5, 4.0, 3.2 and 3.0 for 1970, 1980, 1993 and 1994, respectively)

cycle, the relative frequency of corals in the smallest size class cycled between ~ 40 and 50% of the total population (Fig. 10a); peak frequency of this size class corresponded to peak abundance of the predator *Acanthaster planci*, as predation caused a transfer of colonies from the larger to smaller size classes. The next smallest size class (21 to 40 cm diam) contained $\sim 30\%$ of the remaining colonies and also cycled with predator abundance. The relative abundance of larger corals was very low, as there was not enough time between the intervals of heavy predation for the population to recover.

Increasing frequency of the predation cycle to 12 and 20 yr resulted in longer intervals of heavy predation as well as longer intervals for recovery. Under this scenario, the model predicted that small colonies would still dominate the population (Fig. 10b, c). Because the intervals between periods of high predation were longer in these simulations, more colonies moved from smaller into larger size classes. Thus, as the period lengthened, the minimum percentage of colonies in the smallest size classes decreased from 40% for the 8 yr simulation to 32 and 25% for the 12 and 20 yr simulations, respectively.

ENSO scenarios

After 100 yr of simulation including multiple ENSO events, the model predicted that the population would be relatively evenly distributed compared to any of the predation scenarios (cf. Figs. 10 and 11). When historical ENSO events were simulated over 100 yr, the 81 to 100 cm size class was the most frequent, although this dominance was reduced through time (Fig. 11a). This simulation also suggested that the largest size class would increase in dominance through time. By Year 100, the model predicted that $>15\%$ of the colonies would be in the largest size class; this was not the case in the field population during any sampling years from 1970 to 1994.

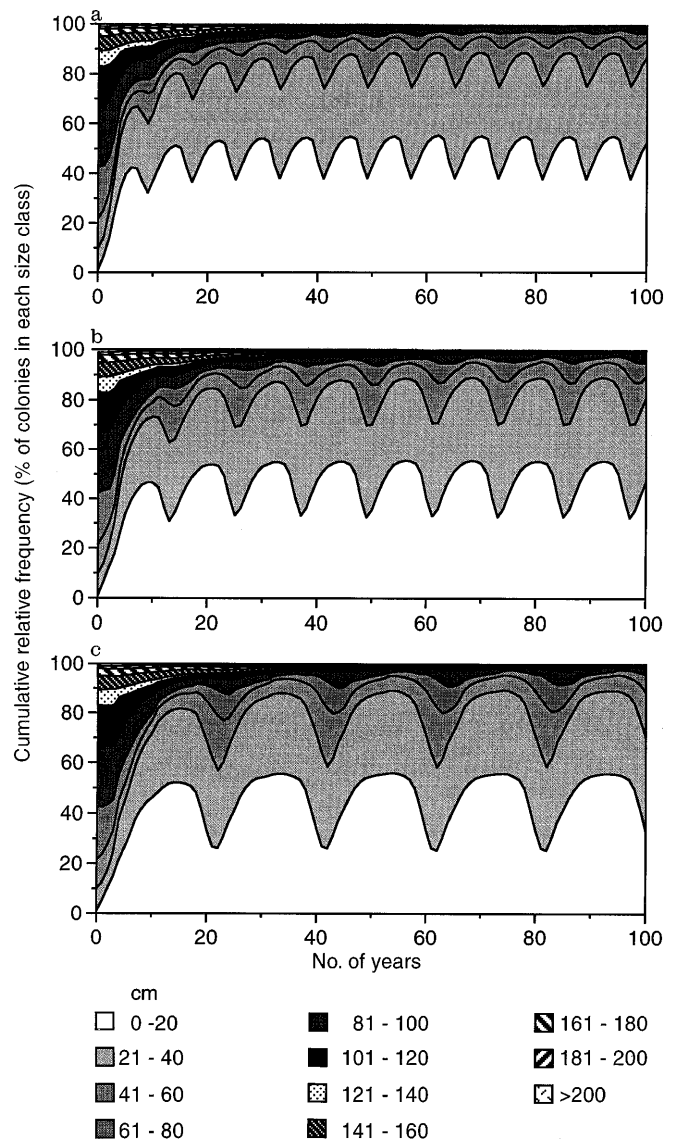


Fig. 10 *Gardineroseris planulata*. Model scenarios predicting changes in size structure of coral population during 100 yr simulation. Size classes are represented by various shadings, and increase in size from bottom to top of graphs. Initial size structure for model population was that of field population on Uva Island reef in 1970. Model conditions were no ENSO activity and an 8 yr (a), 12 yr (b), and 20 yr (c) cyclic pattern of density of predator *Acanthaster planci*

Predictions of the size structure of the population for a recurrence interval drawn from a uniform distribution were not very different from the scenarios based on real data (Fig. 11). In ENSO years, there was a net movement of corals into smaller size classes; however, without predation, recovery always occurred in ~ 2 yr. Results of the other alternative scenario were similar; although recurrence intervals were different, transitions into smaller size classes were small and recovery was rapid.

Additional simulations that tested for long-term effects showed that a mean recurrence interval of 3 yr (SD = 2) resulted in more frequent changes in the size structure of the population, and changes were of greater

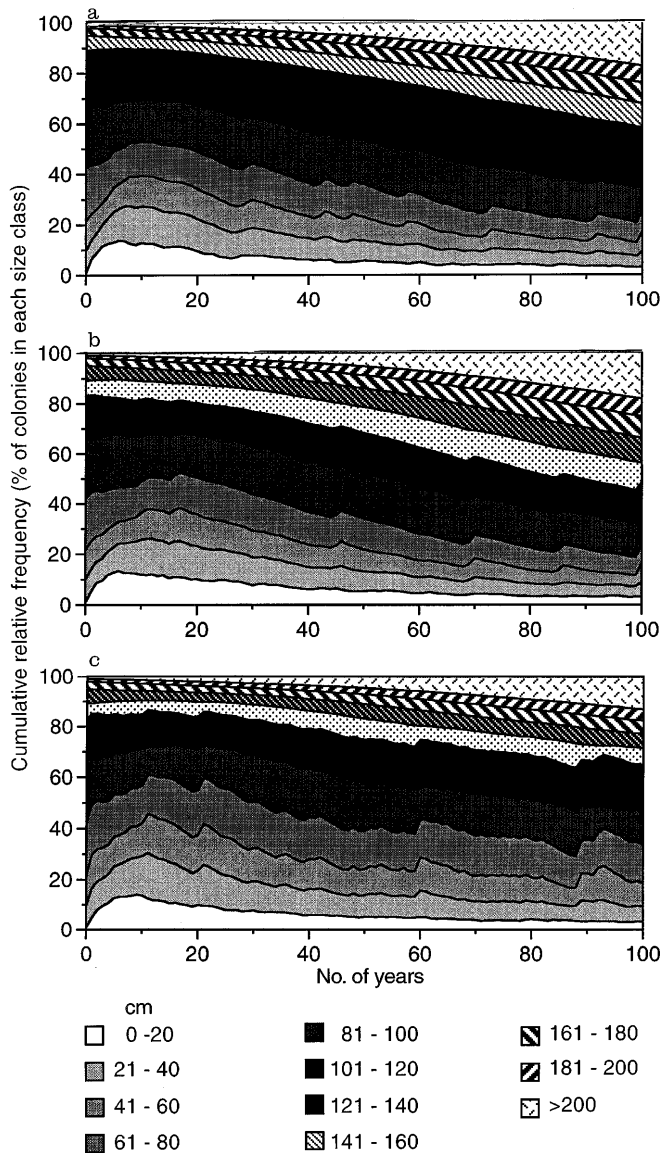


Fig. 11 *Gardineroseris planulata*. 100 yr model scenarios with no predation. Model runs include ENSO activity under three formulations: **a** conditions of actual ENSO activities in strong to very-strong categories from historical records (Glynn 1988); **b** ENSO recurrence times drawn randomly from uniform distribution ranging from 0 to 24; **c** ENSO recurrence times drawn from normal distribution, with mean interval between ENSO events = 11.95 yr (SD = 6.60). Size classes are represented by various shadings, and increase in size from bottom to top of graphs

magnitude than those in the more realistic scenarios (Fig. 11c). In addition, the model predicted that under a more frequent ENSO regime, colonies >120 cm would be less frequent.

Combining ENSO and predation scenarios

When simulations combining ENSO and all three predation scenarios were run, model predictions were not visually distinguishable from those generated for pre-

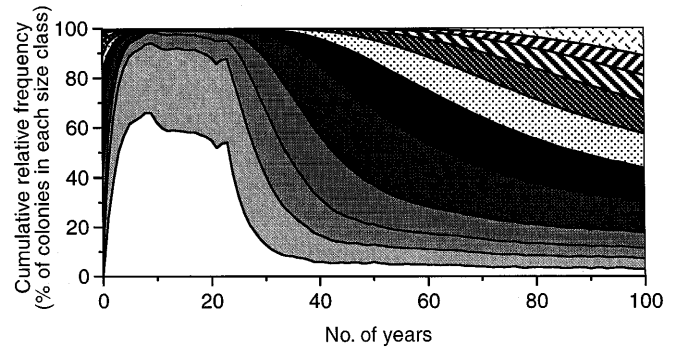


Fig. 12 *Gardineroseris planulata*. 100 yr model scenario run under Uva Island conditions from 1970 to 1994 as Years 0 to 24, and then run for the next 76 yr with both predation and ENSO sectors of model set to zero. (Shadings as in Figs. 10 and 11)

dation alone. This result suggested that, for *Gardineroseris planulata* at Uva Island reef, predation was of overwhelming importance to population size-structure.

Investigating population recovery

In the simulation where disturbance was simulated then removed (Fig. 12), the original size structure of the population (shown in Fig. 8a) was re-established by Year 55, 31 yr after the cessation of disturbance. By this year, each of the size classes was within 1% of the original size structure. The 81 to 100 cm size class was the most frequent, followed by the 101 to 120 cm and the 61 to 80 cm size classes. As in the 1970 population, the least-frequent classes were the smallest and largest. However, by the end of the 100 yr simulation, the population was skewed toward the larger size classes, a situation never measured for the field populations.

Discussion

One important contribution simulation-modeling can make toward any research effort is to synthesize accumulated field and experimental data. However, perhaps more importantly, modeling can be a useful tool for developing hypotheses to aid in future research. Results of the simulations presented in this paper suggested some interesting comparisons between the effects of the two kinds of disturbances, predation and ENSO, that can be tested in the field or laboratory.

The results of the model simulations suggested that predation was far more important than ENSO in controlling the size structure of *Gardineroseris planulata* in the eastern Pacific. On Uva Island reef, abundances of predators were always low, well below the hundreds (Ormond and Campbell 1974) to thousands ha^{-1} (Moran et al. 1985) considered to be "outbreak" levels elsewhere. Endean (1973, 1976) found that at low population densities of *Acanthaster planci*, destruction to a

reef was negligible and damage was quickly healed. Similarly, for a mixed coral community in the western Pacific, Keesing and Lucas (1992) estimated that there was a threshold level of 10 sea stars ha^{-1} above which depletion of live coral cover occurred. In contrast, a model based on *A. plani* predation rates at the Uva Island coral community indicated that reef growth would remain positive at *A. plani* densities of up to 65 sea stars ha^{-1} (Glynn 1973). However, this was net growth of a reef dominated by *Pocillopora* spp. Model results and field data presented here showed that changes in the population size-structure of preferred prey species such as *G. planulata* occurred at lower densities. From these model results we hypothesized that individuals in the real world would achieve large size only in special circumstances, i.e. if they were protected by a biotic barrier, if they were in cryptic locations, or if they were in areas from which *A. plani* were absent (i.e. upwelling environments).

There were both similarities and differences between the results of our model and the results of a model simulating the effects of *Acanthaster planci* predation on the abundance and size structure of *Porites* spp. in the Western Pacific (Done 1987, 1988). Both models found predation to be a very important factor controlling coral population-dynamics. However, the patterns of predation and the effects on the coral populations were very different in the two regions. Done modeled the effects of periodic outbreaks of predators. During each outbreak, predators killed up to 25% of the population and reduced the size of almost all colonies; however, between outbreaks, the effects of predators were negligible. On Uva Island reef, predators occurred in low levels almost every year. For *Porites* spp. in the Western Pacific, model simulations using realistic schedules of survivorship and recruitment suggested that disturbance intervals of ≥ 30 yr were necessary for the population to maintain colonies in the largest size class; the interval between recent outbreaks was 15 yr. Our model suggested that the abundances of predators measured in the past 24 yr would not allow the population to maintain colonies in the largest size class; thus, our model, like Done's, also suggested that abundances measured in the recent past may not be typical of historical abundances. Interestingly, recovery times after the cessation of disturbance for each of these populations were comparable, 55 yr for *Gardineroseris planulata* and 50 to 100 yr for *Porites* spp. (Done 1987).

Results of model simulations also suggested that, for this species of coral, mortality due to ENSO was not very important. Simulations demonstrated that ENSO-related mortality was detectable only in the absence of predation. However, it is possible that in areas where predation is absent, ENSO-related mortality may become important. A comparison of model predictions with the size-frequency distribution of a coral population in an upwelling environment would test the hypothesis that ENSO is important in the absence of *Acanthaster planci*.

There are at least two possible reasons that predation was more important than ENSO in controlling coral population-dynamics. First, transition rates due to predation were both larger and occurred more frequently than rates due to ENSO. Predation occurred every year, while ENSO occurred periodically. However, simulations varying frequency of ENSO demonstrated that the growth rates incorporated into the model were able to keep pace with ENSO recurrence times of ≥ 3 yr. In contrast, even the least-frequent *Acanthaster planci* cycle of abundance resulted in a population size-structure skewed toward small individuals because predation driven transitions were larger than the transitions due to growth. Second, ENSO events seemed to affect all size classes similarly. In contrast, predators concentrated on a few size classes, and consequently had a strong enough effect on them to change the shape of the distribution as a whole.

Although, in general, the simulation-model predictions matched field observations of the size structure of the coral population, there were a few important discrepancies. When simulating the Uva Island population over 24 yr, *Acanthaster planci* predation effects were slightly underestimated. Although this did not cause a statistically significant difference between the size structure of the modeled and real population over a 24 yr period, it may be of concern over a longer period. Predation is a repeated, yearly disturbance, and small discrepancies can compound over time. Thus, improving our estimate of the effect of predation is of key importance to refine or expand the usefulness of the model. The same argument applies to the recruitment portion of the model. Although variances were low over relatively short periods of time, these errors can compound over the longer term.

Comparison of model predictions to field data also revealed that there were very low levels of unexplained mortality in the absence of the predator. Although in years with *Acanthaster planci* this small amount of mortality was masked by the larger effect of predation, in years without predation or areas where predators are absent, this discrepancy may become more important. Therefore, it is important that future research focuses on other sources of mortality that are not included in this model.

Our future efforts will focus on obtaining better estimates of growth, recruitment, and predation rates for the present model. We have stained colonies in *Gardineroseris* City that, in time, will provide a more realistic measure of temporal variability in growth. Additionally, testing the model against field data on population size-structure collected in areas without predators or differing predator abundances will test the hypotheses developed here on the relative importance of predation and ENSO. Finally, we will parameterize this model for other species of coral. *G. planulata* colonies are not as sensitive to high temperatures as other species of coral, and are preferred prey for *Acanthaster planci*. Comparing the importance of these two factors will be

very different when heat-sensitive *Pocillopora* spp. populations that are rarely preyed upon are simulated.

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